Project Title- Loan Approval Prediction

Objective

The objective of this project is to analyze a loan application dataset and build a machine learning model that predicts whether a loan application will be approved based on applicant details such as income, education, credit history, and loan amount.

Dataset

The dataset contains applicant information including Gender, Married status, Dependents, Education, Self-Employment, Applicant Income, Coapplicant Income, Loan Amount, Loan Amount Term, Credit History, Property Area, and Loan Status.

Data Preprocessing

- Loaded the dataset into Google Colab using Pandas.
- Checked the dataset structure using `df.info ()` and `df.head()`.
- Identified missing values using `df.isnull().sum()`.
- Filled missing numerical values using the column mean.
- Prepared the dataset for machine learning by separating the input features and target variable.

Visualizations

The following visualizations were created:

- Loan Approval Status Count Plot
- Applicant Income Distribution
- Credit History Vs Loan Status
- Correlation Heatmap

These visualizations helped understand the relationship between applicant details and loan approval.

Machine Learning Model

A **Decision Tree Classifier** was used to predict loan approval.

The dataset was split into training and testing sets using an 80:20 ratios. The model was trained on the training data and evaluated using prediction accuracy on the testing data.

Model Accuracy: 71.54 %

Key Findings

- Applicants with a good credit history have a higher chance of loan approval.
- Applicant income positively influences loan approval.
- Loan amount and repayment term also affect approval decisions.
- Data preprocessing improved the quality of the dataset before model training.

Conclusion

The Loan Approval Prediction model successfully predicts whether a loan application is likely to be approved using applicant information. This project demonstrates the application of data preprocessing, exploratory data analysis, visualization, and machine learning techniques for solving a real-world financial prediction problem.

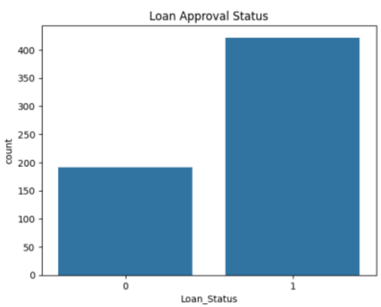

Fig1-Loan approval

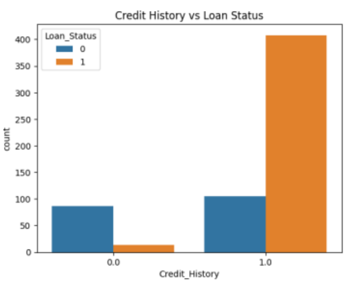

Fig-2 Credit History

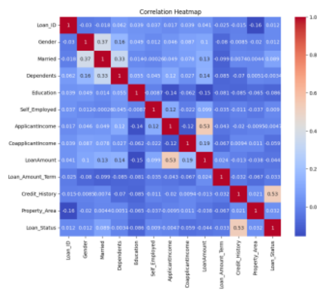

Figure3- Heatmap